UNILAG Admission Compass

Product Requirements Document

Written as job stories

A web tool that helps UNILAG aspirants assess their admission chances before sitting for Post-UTME.

1. The Backdrop

Every year, tens of thousands of Nigerian secondary school graduates apply to the University of Lagos. Between writing UTME and sitting Post-UTME, they sit in an information vacuum. They have a JAMB score, they have O'Level results, they know their dream course — but they have no clear way to answer the question that matters most: am I going to get in?

UNILAG publishes everything needed to answer that question — the cut-off marks for every department, the subject combinations required for each course, the formula for combining UTME, O'Level, and Post-UTME into a 100-point aggregate. But pulling it all together yourself requires understanding the math, knowing your subject combination is valid, and figuring out whether you qualify under merit or catchment.

UNILAG Admission Compass closes that gap. The aspirant inputs their name, target department, UTME score, O'Level grades, and catchment status. The app delivers a personalized, plain-English answer in seconds: where they stand, what Post-UTME score they need, how their catchment status helps, and which alternative departments they already comfortably qualify for.

How this PRD is structured

This document is written as job stories rather than feature lists. Every requirement starts with a real moment in the aspirant's life, framed as: "When I [situation], I want to [motivation], so I can [outcome]." Each story is followed by acceptance criteria — concrete, testable conditions that must be true for the story to count as done.

The job-story format keeps us anchored in why each piece exists. If a feature can't be expressed as a real moment in someone's day, it probably doesn't belong in v1.

2. Who We're Building For

Three kinds of people will use this app. Designing for the first one is non-negotiable; the other two are bonuses we get for free if we get the first one right.

The aspirant

A Nigerian secondary school graduate, sixteen to twenty years old, who has written UTME, has their O'Level results, and is preparing for the UNILAG Post-UTME. Mobile-first — almost certainly using a phone, often on a metered cellular connection. Discovers tools like this through WhatsApp groups, Twitter, school forums.

The parent

The parent or older relative of an aspirant. Wants the official-looking version of the result to forward in family group chats. Less interested in flexibility, more interested in legitimacy.

The counsellor

A secondary-school counsellor or tutorial-centre instructor advising a cohort of aspirants. Uses the alternative-departments feature as a counselling tool when a student's first choice is out of reach.

3. The Job Stories

This is the heart of the PRD. Each story below describes a real situation an aspirant finds themselves in, what they want to accomplish, and why. The acceptance criteria below each story define what "done" looks like for the developer.

Story 1: A quick reality check

When I'm sitting at home a few weeks after writing UTME, scrolling on my phone with my JAMB result and WAEC slip in front of me,

I want to type my scores into a simple form and instantly see whether my dream UNILAG course is realistic for me,

So I can stop second-guessing and start preparing for Post-UTME with a clear target.

Acceptance criteria:

- The form fits a single column on a 360px-wide phone screen with thumb-friendly tap targets (minimum 48px height).
- The user can complete the form in under 60 seconds with their results in hand.
- After submitting, the result appears inline on the same page (no navigation), so the user sees their answer immediately.
- No login, no sign-up, no email required — the user can use the app the moment they land on it.

Story 2: Picking the right O'Level subjects

When I'm filling in my O'Level grades and I'm not sure which five subjects UNILAG actually counts for my chosen department,

I want to have the app show me exactly which subjects to enter — with English and Maths locked in by default and the other three either fixed or chosen from a valid shortlist,

So I can trust that the calculation reflects what UNILAG would actually use, not a guess on my part.

Acceptance criteria:

- The moment the user selects a department, the O'Level subject section reconfigures to match that department's official requirements.
- English Language and Mathematics appear as locked subject rows (the user cannot change them) for every department.
- Course-specific required subjects appear as additional locked rows (e.g. Biology, Chemistry, Physics for Medicine).

- Where UNILAG allows a choice (e.g. Computer Science accepts Chemistry OR Biology as the fifth subject), the app shows a dropdown limited to the valid options for that elective slot.
- Subject rules are sourced strictly from UNILAG's official 2025/2026 admission requirements PDF — no guessing, no inferred substitutions.

Story 3: Knowing my catchment advantage

When I'm from Lagos and I've heard catchment students get in easier, but I don't know exactly how much that helps me,

I want to see in plain numbers how my catchment status changes my required Post-UTME score compared to a non-catchment applicant,

So I can feel motivated and stop underestimating my chances.

Acceptance criteria:

- The form includes a clear toggle: "Are you from a UNILAG catchment state? (Lagos, Ogun, Oyo, Osun, Ondo, Ekiti)"
- If the user selects Yes, a state dropdown appears with the six valid options.
- The result calculation uses the catchment cut-off (which is typically lower) instead of the merit cut-off.
- The result page shows a clearly labeled "Catchment boost" block: "As a Lagos indigene, your cut-off drops from 85.025 to 79.75. That saves you 5.275 points — your Post-UTME target moves from 30/30 to 24.7/30."
- If the catchment cut-off would change the user's verdict tier (e.g. from "Plan B time" to "Lock in"), the boost block is given visual prominence.
- If UNILAG published "NIL" for the user's state-department combo, the app falls back to the merit cut-off and explicitly tells the user "No catchment cut-off was published for [state] in this department; the merit cut-off applies."

Story 4: A verdict in plain English, not jargon

When I get my result back and I just want to know whether I'm in good shape or in trouble,

I want to see a verdict written like a real person would say it — with my name, the actual number I need to score, and a clear instruction on what to do next,

So I can skip the mental gymnastics and either celebrate, get serious, or pivot to a Plan B.

Acceptance criteria:

- The verdict headline uses one of six traffic-light verbs based on required Post-UTME score (see table below).
- The headline addresses the user by name (e.g. "Lock in, Yasir.").

- The body text always includes the actual required Post-UTME number, rounded to one decimal place (e.g. "You need 24.7 out of 30 in Post-UTME").
- Every verdict mentions the 12-mark disqualification rule: "Anything below 12 is automatic disqualification regardless of how well you scored in JAMB."
- The result never uses abstract jargon like "Tier 3 Competitiveness" or "Achievable" in user-facing copy — only the plain-English headlines and body text shown below.

Verdict mapping (used internally; never shown as labels):

Required PUTME	Headline	Body sentence
<= 12	▯ You're in.	As long as you don't score below 12 in Post-UTME (which is automatic disqualification), you're getting admitted.
13-18	▯ You're in.	You need [X]/30 in Post-UTME — well within reach with steady prep.
19-23	▯ Push hard.	You need [X]/30 in Post-UTME. That's around average — solid prep gets you there.
24-27	▯ Lock in.	You need [X]/30 in Post-UTME. That's the top end — serious work, not casual prep.
28-30	▯ Top performer.	You need [X]/30 in Post-UTME. Only the very best hit this — but it's still possible.
> 30	⛔Plan B time.	Last year's cut-off would require more than the maximum Post-UTME score. Take a look at the alternative departments below.

Story 5: Knowing how hard the course itself is

When I'm picking a department but I don't know if I'm aiming at one of the toughest courses or one of the easier ones,

I want to see a one-line note about how competitive that course is at UNILAG before I see my personal result,

So I can calibrate my expectations and not be blindsided by my own chances.

Acceptance criteria:

- Above the personal verdict, the result page shows a course-difficulty sentence based on the department's merit cut-off.
- The sentence uses one of five plain-English phrases: "Heads up — this is one of the hardest courses to get into at UNILAG" (cut-off >=80), "This is a highly competitive course at UNILAG" (70-79.99), "This is a moderately competitive course at UNILAG" (60-69.99), "This is one of the more accessible UNILAG courses" (50-59.99), or "This course has a relatively low admission barrier at UNILAG" (< 50).

- The sentence is followed by the actual cut-off number for transparency: "Last year's cut-off was 85/100."

Story 6: Finding a realistic backup plan

When I just got the bad news that my dream course is out of reach this year,

I want to see three other UNILAG departments where I'm already comfortably positioned with my current scores,

So I can have a serious Plan B I can actually qualify for instead of feeling completely stuck.

Acceptance criteria:

- After the verdict, the app shows up to three alternative departments where the user's verdict would be "You're in" (required Post-UTME <= 18).
- Suggested departments must be ones the user's current O'Level subjects qualify them for (a science candidate isn't shown an arts course they can't enter).
- Suggestions are ranked by merit cut-off (highest first), so the user sees the most prestigious accessible options first.
- Each suggestion shows: department name, faculty, required Post-UTME, and a one-tap "Try this course" button that recalculates with the new department.
- If the user has fewer than three qualifying alternatives, the app shows whatever is available (one or two), or a friendly message: "We didn't find easier alternatives that match your subjects. Consider broadening your subject combination."

Story 7: Sharing my result

When I've just gotten my result and I want to share it with my friends in our WhatsApp aspirants group,

I want to tap a single button and get a clean, branded image of my result that I can drop straight into a chat or status,

So I can show off, get feedback, and indirectly bring more friends to the tool.

Acceptance criteria:

- A primary call-to-action button labeled "Share my result" is visible on the result page without scrolling on a 360px viewport.
- Tapping it opens a modal with two tabs: "Professional" and "Social".
- The Professional card is portrait, white background, UNILAG-blue accent, structured like a mini transcript: name, department, current aggregate, applicable cut-off, required Post-UTME, verdict.
- The Social card is portrait at 9:16 ratio (1080x1920), with a high-contrast gradient background, oversized verdict headline, and is clearly designed for WhatsApp Status / Instagram Stories.
- Both cards are generated client-side from the result data using html-to-image (no server round-trip).

- Each tab provides a Download button (saves as PNG) and a Share button that uses the Web Share API on supported devices, falling back to download elsewhere.
- Both cards include a small footer attributing the tool and a short URL, plus a disclaimer line: "Based on UNILAG 2025/2026 official data — your year may differ."

Story 8: Comparing different courses

When I want to see how my chances differ across two or three related courses (say Medicine, Pharmacology, and Nursing),

I want to change just the department on my form and instantly recalculate without re-entering my UTME score and grades,

So I can make a more informed choice about which course to actually apply for.

Acceptance criteria:

- From the result view, a secondary button labeled "Try a different course" is always visible.
- Tapping it scrolls back to the form with all inputs (name, UTME score, O'Level grades, catchment status) preserved.
- Only the department and the elective subject slots reset (because they depend on the new department).
- The result re-renders the moment the user resubmits with the new department.
- Form state is preserved using the Storage abstraction so the calculation persists across page refreshes within the same browser session.

Story 9: Catching disqualification early

When I'm filling out the form with a UTME score below 200, or with O'Level grades that include a fail,

I want to see a clear, friendly warning before I get a misleading result,

So I can understand the actual UNILAG rules and not waste hope on an application that will be rejected outright.

Acceptance criteria:

- If the user enters a UTME score below 200, the app blocks calculation and shows: "JAMB requires a minimum UTME score of 200 for admission to any UNILAG programme. You'll need to retake UTME to be eligible."
- If any O'Level grade is below C6 (i.e. D7, E8, or F9), the app blocks calculation and shows: "UNILAG requires a credit pass (C6 or above) in all five required O'Level subjects. The grade for [subject] doesn't meet this requirement."
- Both warnings are presented inline near the offending field, not as a modal pop-up.

- The Post-UTME 12-mark disqualification rule is always shown alongside the result (even when the calculation succeeds), so the user understands it before sitting Post-UTME.

Story 10: Knowing this isn't a guess

When I'm reading my result and starting to make real decisions based on it,

I want to see clear evidence that the numbers come from UNILAG itself, not someone's blog,

So I can trust the result enough to act on it.

Acceptance criteria:

- The footer of every page shows: "Based on UNILAG's official 2025/2026 cut-off marks and admission requirements" with direct links to both source documents.
- The About page contains the full aggregate formula explainer, the data-year disclaimer, and links to UNILAG's official cut-off page and admission requirements PDF.
- The About page makes it explicit that cut-offs change yearly and the tool reflects last year's data only.
- Both share-card variants include the data-year footer so screenshots passed around still reflect the source.

4. What's Not in v1

These are deliberately excluded from the first version. Some are nice ideas that just don't fit the timeline; some are bigger projects that deserve their own PRD.

- Direct Entry (DE) admission flow — v1 covers UTME applicants only.
- User accounts, login, password reset, and any kind of authentication.
- Server-side data persistence or any database. All state lives in the browser.
- Predictions about future cut-offs. The tool only compares against last published year.
- Real-time integration with JAMB CAPS, UNILAG admission portals, or any external API.
- Postgraduate, foundation, distance learning, or ICE programme support.
- ELDS-specific calculations. UNILAG doesn't publish ELDS cut-offs in the same table; we acknowledge ELDS exists but don't compute against it.
- Departments that show NIL across all categories in UNILAG's release (no admissions data) — excluded from search until UNILAG publishes a number.

5. The Numbers (Formula and Rules)

Every calculation in the app rests on UNILAG's published aggregate formula. This section is the source of truth for the developer.

5.1 The aggregate

Component	Calculation	Max points
UTME (JAMB)	Raw score divided by 8	50
O'Level	Sum of point values for 5 required subjects	20
Post-UTME	Raw score (already on a 30-point scale)	30
Total	Sum of the three components	**100**

5.2 O'Level grade points

Credit passes are A1=4.0, B2=3.6, B3=3.2, C4=2.8, C5=2.4, C6=2.0. Grades D7, E8, and F9 are failing grades and disqualify the candidate from using that subject. The user must produce five credit passes — anything else blocks calculation entirely.

5.3 The reverse calculation

The whole point of the app. Given the target department's applicable cut-off and the user's UTME and O'Level inputs, compute the Post-UTME score the user must hit:

```
requiredPostUTME = applicableCutoff - (utmeScore / 8) - sum(oLevelPoints)
```

5.4 Hard rules to enforce

- UTME below 200: blocks calculation. JAMB minimum applies.
- Post-UTME below 12: automatic disqualification regardless of UTME score. Always shown alongside the result.
- Fewer than five credit passes: blocks calculation. Five credit passes are required.
- English Language and Mathematics are compulsory for every department.
- If applicableCutoff − currentAggregate is negative, the user is already past the cut-off; render the verdict as "You're in" with a friendly note that they only need to clear the 12-mark Post-UTME minimum.

6. The Domain Model

The calculation logic is encapsulated in an object-oriented domain layer written in TypeScript. The model uses abstract base classes, enums, and inheritance to isolate logic from the UI, make the codebase trivial to test, and create clean extension points for v2.

6.1 Enums

```
enum Faculty {
  ARTS = "Faculty of Arts",
  BASIC_MEDICAL_SCIENCES = "Faculty of Basic Medical Sciences",
  CLINICAL_SCIENCES = "Faculty of Clinical Sciences",
  DENTAL_SCIENCES = "Faculty of Dental Sciences",
  EDUCATION = "Faculty of Education",
  ENGINEERING = "Faculty of Engineering",
  ENVIRONMENTAL_SCIENCES = "Faculty of Environmental Sciences",
  LAW = "Faculty of Law",
  MANAGEMENT_SCIENCES = "Faculty of Management Sciences",
  PHARMACY = "Faculty of Pharmacy",
  SCIENCE = "Faculty of Science",
  SOCIAL_SCIENCES = "Faculty of Social Sciences",
}

enum OLevelGrade { A1, B2, B3, C4, C5, C6, D7, E8, F9 }
enum CatchmentState { EKITI, LAGOS, OGUN, ONDO, OSUN, OYO }
enum AdmissionCategory { MERIT, CATCHMENT }
enum SubjectCategory { CORE, SCIENCE, ARTS, SOCIAL_SCIENCE, LANGUAGE,
TECHNICAL }

enum PersonalVerdict {
  YOURE_IN_COMFORTABLE,   // <= 12
  YOURE_IN_STRONG,        // 13-18
  PUSH_HARD,              // 19-23
  LOCK_IN,                // 24-27
  TOP_PERFORMER,          // 28-30
  PLAN_B_TIME             // > 30
}

enum CourseDifficulty {
  EXTREMELY_HARD,         // >= 80
  HIGHLY_COMPETITIVE,     // 70-79.99
  MODERATELY_COMPETITIVE, // 60-69.99
  ACCESSIBLE,             // 50-59.99
  LOW_BARRIER             // < 50
}
```

6.2 Abstract base classes

```
abstract class Scoreable {
  abstract getScore(): number;
  abstract getMaxScore(): number;
  getPercentage(): number {
    return (this.getScore() / this.getMaxScore()) * 100;
  }
}

abstract class VerdictBearing<T> {
  abstract getVerdict(): T;
  abstract getHeadline(): string;
  abstract getBody(): string;
  abstract getEmoji(): string;
}
```

6.3 Subject hierarchy

```
abstract class Subject {
  constructor(
    public readonly name: string,
    public readonly category: SubjectCategory,
    public readonly aliases: string[] = []
  ) {}
  abstract isCompulsoryFor(department: Department): boolean;
}

class CoreSubject extends Subject {
  // English & Mathematics — compulsory for ALL departments
  isCompulsoryFor(_dept: Department): boolean { return true; }
}

class CourseSpecificSubject extends Subject {
  isCompulsoryFor(dept: Department): boolean {
    return dept.requiredSubjects.includes(this.name);
  }
}

class ElectiveSubject extends Subject {
  isCompulsoryFor(_dept: Department): boolean { return false; }
}
```

6.4 Score classes

```
abstract class ExamScore extends Scoreable {
  constructor(public readonly raw: number) { super(); }
```

```
}

class UTMEScore extends ExamScore {
  getScore(): number { return this.raw / 8; }
  getMaxScore(): number { return 50; }
  meetsMinimum(): boolean { return this.raw >= 200; }
}

class PostUTMEScore extends ExamScore {
  getScore(): number { return this.raw; }
  getMaxScore(): number { return 30; }
  isDisqualified(): boolean { return this.raw < 12; }
}

class Grade extends Scoreable {
  private static readonly POINTS: Record<OLevelGrade, number> = {
    A1: 4.0, B2: 3.6, B3: 3.2, C4: 2.8, C5: 2.4, C6: 2.0,
    D7: 0, E8: 0, F9: 0,
  };
  constructor(public readonly value: OLevelGrade) { super(); }
  getScore(): number { return Grade.POINTS[this.value]; }
  getMaxScore(): number { return 4.0; }
  isCredit(): boolean { return this.getScore() >= 2.0; }
}

class OLevelResult extends Scoreable {
  constructor(public readonly results: SubjectResult[]) {
    super();
    if (results.length !== 5) {
      throw new Error("UNILAG requires exactly 5 O'Level subjects");
    }
  }
  getScore(): number {
    return this.results.reduce((s, r) => s + r.grade.getScore(), 0);
  }
  getMaxScore(): number { return 20; }
  hasAllCredits(): boolean {
    return this.results.every(r => r.grade.isCredit());
  }
}
```

6.5 Department, Candidate, Assessment

```
interface CutoffSet {
  merit: number;
  catchment: Partial<Record<CatchmentState, number | null>>;
}
```

```
interface ElectiveGroup {
  pickCount: number;
  options: string[];
}

interface SubjectRequirement {
  required: string[];
  electiveGroups: ElectiveGroup[];
}

class Department extends VerdictBearing<CourseDifficulty> {
  constructor(
    public readonly name: string,
    public readonly faculty: Faculty,
    public readonly cutoffs: CutoffSet,
    public readonly subjectRequirement: SubjectRequirement
  ) { super(); }

  getCutoffFor(category: AdmissionCategory, state?: CatchmentState): number {
    if (category === AdmissionCategory.CATCHMENT && state) {
      return this.cutoffs.catchment[state] ?? this.cutoffs.merit;
    }
    return this.cutoffs.merit;
  }

  getVerdict(): CourseDifficulty { /* logic from cutoffs.merit */ }
  getHeadline(): string { /* "This is a highly competitive course at UNILAG."
*/ }
  getBody(): string { /* "Last year's cut-off was 78.225/100." */ }
  getEmoji(): string { /* □ ⚡ □ ✅□ */ }
}

class Candidate {
  constructor(
    public readonly name: string,
    public readonly utme: UTMEScore,
    public readonly oLevel: OLevelResult,
    public readonly catchmentState: CatchmentState | null,
    public readonly targetDepartment: Department
  ) {}
  isCatchment(): boolean { return this.catchmentState !== null; }
  getCurrentAggregate(): number {
    return this.utme.getScore() + this.oLevel.getScore();
  }
  getApplicableCategory(): AdmissionCategory {
    return this.isCatchment()
      ? AdmissionCategory.CATCHMENT
```

```
      : AdmissionCategory.MERIT;
  }
}

class AdmissionAssessment extends VerdictBearing<PersonalVerdict> {
  public readonly currentAggregate: number;
  public readonly applicableCutoff: number;
  public readonly meritCutoff: number;
  public readonly requiredPostUTME: number;
  public readonly catchmentBoost: number;

  constructor(public readonly candidate: Candidate) {
    super();
    const dept = candidate.targetDepartment;
    this.currentAggregate = candidate.getCurrentAggregate();
    this.meritCutoff = dept.cutoffs.merit;
    this.applicableCutoff = dept.getCutoffFor(
      candidate.getApplicableCategory(),
      candidate.catchmentState ?? undefined
    );
    this.requiredPostUTME = this.applicableCutoff - this.currentAggregate;
    this.catchmentBoost = this.meritCutoff - this.applicableCutoff;
  }

  getVerdict(): PersonalVerdict { /* mapping from requiredPostUTME */ }
  getHeadline(): string { /* "Lock in, Yasir." */ }
  getBody(): string { /* full plain-English sentence */ }
  getEmoji(): string { /* □ □ □ □ ⛔ */ }
}
```

7. Data Sources

Two JSON files drive the entire app. Both are updated annually when UNILAG releases new admission data.

7.1 cutoffs.json

Sourced from UNILAG's official cut-off announcement page. The v1 source is the 2025/2026 release at unilag.edu.ng/unilag-releases-2025-2026-utme-merit-cut-off-marks. Each entry stores the department name, faculty, merit cut-off, and a catchment object keyed by state. NIL values in the source become null in the JSON and the calculator falls back to merit.

7.2 requirements.json

Sourced from UNILAG's official admission requirements PDF at admissions.unilag.edu.ng/assets/file/2025-2026_Admission_Requirements.pdf. Each entry contains the SubjectRequirement structure (required subjects + elective groups). The data-extraction is a one-time manual job per year — roughly two hours for the full ~80-department list.

7.3 Annual update workflow

Each October when UNILAG publishes new cut-offs, the maintainer (1) updates cutoffs.json from the new web release, (2) updates requirements.json from the new PDF if changes were made, (3) runs the test suite to verify nothing breaks, (4) bumps the version label visible in the app footer.

8. Technical Architecture

8.1 Stack

- Next.js 15 with the App Router and TypeScript.
- Tailwind CSS v4 for styling.
- shadcn/ui for component primitives.
- html-to-image for share-card PNG generation.
- Browser localStorage for persisting the user's last calculation across page refreshes.
- Vercel hobby tier for hosting.
- No backend, no database, no API calls beyond initial page load.

8.2 Project layout

```
unilag-compass/
├── app/
│   ├── page.tsx                    # Landing + form + result inline
│   ├── about/page.tsx              # FAQ + disclaimer + formula explainer
│   └── layout.tsx
├── components/
│   ├── form/
│   │   ├── AspirantForm.tsx
│   │   ├── DepartmentSelect.tsx
│   │   ├── OLevelInputGrid.tsx
│   │   └── CatchmentToggle.tsx
│   ├── result/
│   │   ├── ResultPanel.tsx
│   │   ├── VerdictBlock.tsx
│   │   ├── CatchmentBoostBlock.tsx
│   │   ├── AlternativesBlock.tsx
│   │   └── ShareModal.tsx
│   ├── share-cards/
│   │   ├── ProfessionalCard.tsx
│   │   └── SocialCard.tsx
│   └── ui/                         # shadcn primitives
├── domain/
│   ├── enums.ts
│   ├── base.ts
│   ├── subject.ts
│   ├── grade.ts
│   ├── score.ts
│   ├── department.ts
```

```
│   ├── candidate.ts
│   ├── assessment.ts
│   └── recommender.ts
├── lib/
│   ├── verdict-copy.ts            # plain-English templates
│   ├── storage.ts                 # localStorage wrapper
│   └── share.ts                   # html-to-image helpers
├── data/
│   ├── cutoffs.json
│   ├── requirements.json
│   └── subjects.json
├── tests/
│   └── domain/                    # Vitest unit tests
└── README.md
```

8.3 Testing

The domain layer is pure — no React, no DOM, no I/O — and is unit-tested with Vitest. The suite covers: the aggregate formula on representative candidates from every faculty, the grade-to-point mapping including failing grades, the disqualification rules at boundary values (UTME exactly 200 vs 199, Post-UTME exactly 12 vs 11), the cut-off lookup including NIL fall-through, the catchment-boost calculation, and the alternatives recommender. UI tests are minimal in v1 — Playwright smoke tests for the happy path and one edge case.

8.4 Performance and accessibility

The home page should load in under 1.5 seconds on a 4G connection on a mid-range Android device, with a Lighthouse mobile Performance score of 90 or higher. Accessibility targets: WCAG AA compliance, fully keyboard-navigable form, semantic HTML, sufficient colour contrast on both share-card variants, screen-reader-friendly result labels.

9. Design Direction

9.1 Visual language

UNILAG's official colour is a deep blue (approximately #003366). Compass uses a complementary palette: deep UNILAG blue as the primary brand colour, an energetic accent (warm gold or coral, suggesting opportunity), and clean greys for supporting type. Typography is a confident geometric sans-serif (Inter or similar) with body text in the same family at a smaller weight.

9.2 Tone of voice

Direct and warm. The app speaks to the user the way an older sibling who's been through admissions would: honest about challenges, encouraging where there's opportunity, never patronising. "You're in" not "Congratulations, your aggregate score satisfies the requirement." "Lock in, Yasir." not "Personal Competitiveness Tier: 4 (Challenging)."

9.3 Mobile-first

Every screen is designed first at 360px width, then scaled up. The form is single-column with generous tap targets (minimum 48px height). The result page is also single-column with card-style blocks separated by clear whitespace. On wider viewports (>=768px) the form and result can render side-by-side, but the mobile flow is the priority.

10. Risks and How We Handle Them

Stale data

If cutoffs.json isn't updated when UNILAG releases new data, the app continues showing last year's numbers. Mitigation: the footer always shows the data year ("Based on UNILAG 2025/2026 official data"), a calendar reminder for October each year, and the about page tells users how to verify against current UNILAG announcements.

Misinterpretation of the verdict

A user might read "You're in" as a guarantee. Mitigation: every verdict body text includes the phrase "based on last year's cut-off", the about page contains a clear disclaimer, and share cards include a "based on 2025/2026 data — your year may differ" footer.

Subject-combination edge cases

UNILAG's PDF contains some unusual phrasings ("any other subject", "two Arts subjects which may include any one of [list]"). Mitigation: any department whose rules can't be cleanly encoded in the SubjectRequirement structure is flagged in the issues tracker, and the app falls back to a permissive subject selector for those edge cases while the maintainer encodes the rule properly.

Maintenance debt

The tool is most valuable during admission season; without updates, it dies in year two. Mitigation: the data layer is intentionally a flat JSON file any contributor can edit without touching code; the README contains a clear annual-update checklist; the calculator is fully tested so updates don't silently break.

11. Launch Acceptance Criteria

The app is ready for v1 launch when all of the following are demonstrably true:

- All ~80 UNILAG UTME degree programmes from the 2025/2026 cut-off release are present in cutoffs.json with both merit and catchment data.
- All corresponding subject requirements are present in requirements.json, validated against the official PDF.
- The aggregate calculator produces correct results on a published worked example, verified by hand-calculation.
- All disqualification rules (UTME below 200, Post-UTME below 12, fewer than five credit passes) trigger appropriate user-facing errors.
- Both share-card variants render correctly on iPhone Safari, Android Chrome, and desktop Chrome.
- Lighthouse mobile Performance score is 90 or higher.
- All Vitest tests in the domain layer pass.
- The about page contains the formula explainer, the data-year disclaimer, and a link to UNILAG's official cut-off page.
- The README contains the annual-update checklist for the maintainer.

— End of PRD —